

COMPUTER TECHNOLOGY

Computer technology is an expanding branch of engineering, with roots in both electrical engineering and computer science. It includes the related areas of computer hardware and software; computer interfacing, programming, and networking; analog and digital electronics; and robotics. Computer technicians, technologists, and engineers work in every sector of society, in careers ranging from building and repairing computer systems to designing and installing computer networks, to designing and building prototype robots and electronic devices.

Students taking computer technology courses will learn to work safely with computer equipment, electronic circuits, and robotic devices, and will explore the Essential Skills and work habits that are important for success in computer technology. Computer technology courses prepare students for apprenticeship, further study at college or university, or entry into the workplace directly after graduation.

The list of approved emphasis areas for computer technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Computer Engineering Technology, Grade 11

University/College Preparation

TEJ3M

This course examines computer systems and control of external devices. Students will assemble computers and small networks by installing and configuring appropriate hardware and software. Students will develop knowledge and skills in electronics, robotics, programming, and networks, and will build systems that use computer programs and interfaces to control and/or respond to external devices. Students will develop an awareness of related environmental and societal issues, and will learn about college and university programs leading to careers in computer technology.

Prerequisite: None

A. COMPUTER TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** describe how computer components function, and discuss trends in the development of computer hardware;
- A2.** describe the functions of BIOSes and operating systems, and how they interact with each other and with computer hardware;
- A3.** describe the function of electronic components and the use of these components in control systems and other circuits, and calculate values for circuit components;
- A4.** describe network concepts, services, and security;
- A5.** demonstrate an understanding of the use of binary numbers, hexadecimal numbers, and Boolean algebra in computer logic and data processing.

SPECIFIC EXPECTATIONS

A1. Computer Hardware

By the end of this course, students will:

- A1.1** describe how the internal components of a computer function (*e.g., CPU, mainboard, disk drives, RAM, chipset, video card, sound card, expansion slot*);
- A1.2** describe various standards for connecting computer components (*e.g., parallel port, RS-232, USB, IEEE 1394, VGA, DVI*);
- A1.3** describe trends in the development of computer hardware (*e.g., size, cost, and speed of processors, memory, and hard drives; video resolution; capacity of optical disks*).

A2. Computer Systems

By the end of this course, students will:

- A2.1** describe the essential functions and other features of various operating systems (*e.g., functions: management of resources, files, processes, and applications; features: services, usability, performance, applications such as text editor, web browser, or media player*);
- A2.2** describe changes that may be required when upgrading hardware components or features of a computer system (*e.g., BIOS updates, installation of drivers for new hardware, resolution of compatibility issues*);

- A2.3** describe the essential functions performed by the BIOS firmware in computer systems (*e.g., POST [power on self test], boot sequence, hardware recognition, detection of master boot record*);

- A2.4** describe how the BIOS, hardware, and operating system of a computer interact.

A3. Electronics, Robotics, and Computer Interfacing

By the end of this course, students will:

- A3.1** identify and describe the functions of electronic components (*e.g., resistor, capacitor, diode, LED*);
- A3.2** describe the function of electrical devices used in control systems (*e.g., stepper motor, direct-current motor, touch sensor, accelerometer, optical sensor, power supply*);
- A3.3** calculate the values of components in electronic circuits using fundamental laws (*e.g., Ohm's law, Kirchhoff's laws*);
- A3.4** explain the importance of advances in electronics (*e.g., compare size, cost, and performance of vacuum tubes, transistors, and integrated circuits*);

A3.5 compare the advantages and disadvantages of interfacing using desktop computers, micro-controllers, and programmable logic controllers.

A4. Networking Concepts

By the end of this course, students will:

- A4.1** explain fundamental network concepts (*e.g., bandwidth, throughput, full duplex, half duplex*);
- A4.2** explain the fundamental aspects of TCP/IP addressing as it pertains to workstations on a network (*e.g., workstation IP address, subnet mask, MAC [media access control] address, default gateway address*);
- A4.3** describe various services offered by servers to network clients (*e.g., HTTP, FTP, SMTP, telnet, printing, file transfers and storage, login*);
- A4.4** describe methods for making a network secure (*e.g., firewalls, data and password encryption, user authentication, WEP or WPA keys, security of server room*).

A5. Data Representation and Digital Logic

By the end of this course, students will:

- A5.1** describe binary and hexadecimal numbers, and convert positive integers among decimal, binary, and hexadecimal number systems;
- A5.2** compare binary and hexadecimal representation of addresses and data (*e.g., absolute addressing, character codes, colours*);
- A5.3** relate Boolean algebra to the fundamental logic gates and to combinations of these gates, using symbolic, algebraic, and numeric representations.

B. COMPUTER TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** build, configure, and maintain a computer system, and connect peripheral devices;
- B2.** set up, optimize, and back up a computer system;
- B3.** design, construct, create diagrams for, and troubleshoot electronic circuits and interfaces for control systems;
- B4.** design, install, configure, test, and troubleshoot networks;
- B5.** demonstrate an understanding of fundamental programming concepts, and develop a program that interacts with an external device.

SPECIFIC EXPECTATIONS

B1. Hardware Solutions

By the end of this course, students will:

- B1.1** build a computer from parts to meet specified requirements (*e.g., for gaming, business, entertainment, media centre, or graphic design*);
- B1.2** use correct procedures to prevent damage to sensitive components (*e.g., use anti-static wrist straps and mats, disconnect power when inserting expansion cards*);
- B1.3** install and configure peripheral devices in a computer system (*e.g., printer, video camera, external drives*);
- B1.4** document maintenance and troubleshooting of computer hardware on a day-to-day basis (*e.g., use a journal or log to record work done, time taken, problems found, solutions attempted, and results*).

B2. Computer Systems

By the end of this course, students will:

- B2.1** set up and configure a home office system (*e.g., computer, scanner, printer, appropriate software*);
- B2.2** use system utilities for optimization and backup (*e.g., defragment files; scan hard drives for defective sectors; run complete, incremental, and differential backups*);
- B2.3** configure a computer system to use multiple operating systems (*e.g., dual boot, virtual machines*).

B3. Electronics, Robotics, and Computer Interfacing

By the end of this course, students will:

- B3.1** use a design process (see pp. 22–23) to design and safely construct and test interfacing or robotics circuits (*e.g., for LED traffic lights, VU meter, alarm system, or motor control*), using appropriate materials and techniques, including soldering;
- B3.2** troubleshoot an electronic circuit using appropriate methods and test equipment (*e.g., methods: isolation and substitution of components; equipment: multimeter, oscilloscope, logic probe*);
- B3.3** draw and interpret diagrams that represent circuit components and functions (*e.g., schematic diagram, block diagram, flow chart*);
- B3.4** use computer programs to simulate circuit performance and to draw schematic diagrams and circuit layouts (*e.g., circuit simulator, schematic capture software, printed circuit board layout software*).

B4. Network Setup and Management

By the end of this course, students will:

- B4.1** design, install, and configure a peer-to-peer network (*e.g., choose appropriate computers and network interfaces, construct cables, enable file sharing*) using appropriate tools, materials, and equipment (*e.g., UTP cable, 8P8C connectors, crimping tool, cable tester*);

- B4.2** draw diagrams of various LAN types (e.g., *peer-to-peer*, *client-server*) and topologies (e.g., *bus*, *star*, *ring*);
- B4.3** construct various network cables (e.g., *straight-through*, *crossover*);
- B4.4** use a variety of methods to verify the operation of a network (e.g., *visual inspection*, *ping*, *ipconfig*, *telnet*, *tracert*, *arp*);
- B4.5** use a problem-solving process (see pp. 21–23) to troubleshoot networks.

B5. Computer Programming

By the end of this course, students will:

- B5.1** use constants, variables, expressions, and assignment statements correctly, taking into account the order in which operations are performed;
- B5.2** use input statements, output statements, selection structures, and repetition structures in a program;
- B5.3** use a design process (see pp. 22–23) to write, test, and debug a computer program that controls and/or responds to the inputs from an external device (e.g., *LED array*, *motor*, *relay*, *infrared sensor*, *temperature sensor*).

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe environmental issues related to the widespread use of computers and associated technologies;
- C2.** describe societal issues related to the widespread use of computers and associated technologies.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the effects of computer and electronic technology on the environment (*e.g., accumulation of electronic waste, including lead and other toxic materials used in computers; release of ozone-destroying chemicals used to wash soldering flux from circuit boards; energy consumed by computers left in standby mode; fuel consumption and air pollution reduced by computerized traffic-control systems*);
- C1.2** outline how community partners and government agencies apply the reduce/reuse/recycle concept to computer technology.

C2. Technology and Society

By the end of this course, students will:

- C2.1** describe the benefits of computer and electronic technology for society (*e.g., greater efficiency and lower costs for information services, improved access to technology for economically disadvantaged people and nations, development of a "global village"*);
- C2.2** describe some of the drawbacks of computer and electronic technology for society (*e.g., loss of privacy, infringement of intellectual property rights through unlicensed copying and electronic distribution, a more sedentary lifestyle, spam, telemarketing, Internet gambling addictions*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of relevant safety practices, standards, and legislation;
- D2.** describe ethical and security issues related to the use of computers;
- D3.** describe various careers related to computer technology and electronics, and the entry requirements for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** comply with relevant industry practices, standards, and related legislation to ensure workplace safety (*e.g., standards and regulations specified in the Workplace Hazardous Materials Information System [WHMIS] and the Electrical Safety Code; grounding and enclosure standards for electrical circuits; ergonomically sound workplace arrangements and practices*);
- D1.2** describe and use appropriate equipment, techniques, and strategies to avoid health and safety problems associated with computer use (*e.g., back injuries from improper lifting of heavy equipment, repetitive strain injuries, eye strain*).

D2. Ethics and Security

By the end of this course, students will:

- D2.1** describe the components of an acceptable-use policy for computers (*e.g., restrictions on commercial or personal use, prohibition of inappropriate content, protection of privacy*);
- D2.2** explain the importance of and comply with software licensing legislation (*e.g., copyright and patent acts*);
- D2.3** explain the importance of security (*e.g., password protection, encryption*) for confidential data and other sensitive electronic information (*e.g., to protect against industrial espionage or identity theft*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** describe various careers related to computer technology and electronics that require

postsecondary education (*e.g., computer engineer, systems analyst, network analyst, information technology technician*);

- D3.2** describe entry requirements, including computer expertise, for careers related to computer technology (*e.g., apprenticeships, university programs, college programs, industry certifications*);

- D3.3** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in computer technology (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

- D3.4** demonstrate an understanding of and apply the Essential Skills that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (OSP) (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);

- D3.5** demonstrate an understanding of and apply the work habits that are important for success in the computer technology industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, organization, working independently, initiative, self-advocacy*);

- D3.6** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in computer technology (*e.g., Passport to Safety certificate, OSP Work Plan, OSP Transition Plan, circuit diagrams, photographs of projects, video of working robot*), and explain why having a current portfolio is important for career development and advancement.